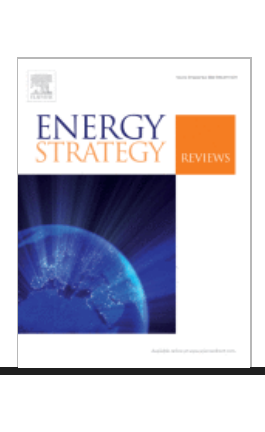

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Optimizing the performance of hybrid renewable energy systems to accelerate a sustainable energy transition in Nigeria: A case study of a rural healthcare centre in Kano

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Highlights

- Many local community health clinics across Africa suffer from energy problems.
- COVID-19 pandemic has brought pressure to the African continent's health sector.
- HOMER was used to optimize different hybrid energy system configurations.
- Combination of RETScreen and HOMER for modelling HES in the health sector.
- RETScreen was used to carry out risk analysis for the PV-Diesel combination.

Abstract

The COVID-19 pandemic brought tremendous pressure to the African continent's health sector, particularly in rural areas. Rural healthcare facilities experienced severe energy shortages due to the influx of Covid-19 patients, which strained their operating capacities. These facilities usually rely on diesel engines, and this unidirectional approach limited governments' vision in resolving the energy crisis. This paper aims to develop a hybrid energy system (HES) that combines alternate energy sources available at a rural healthcare centre in northern Nigeria and conduct a techno-economic analysis of the viability of such systems. The energy potential of available energy resources at the target location and comparison of various HES configurations were estimated with HOMER Pro software to determine the most feasible choices. These choices were analyzed for technical, financial, and environmental viability using RETScreen Expert. Due to an abundance of solar energy, low diesel cost, and sufficient wind energy potential, the results show PV – Diesel and Wind – Diesel system HES configurations as the only practical choices. Techno-economic analysis indicates that the PV – Diesel HES produces annual savings of $30,583 with a Net Present Value (NPV) of $390,949 compared to $15,174 and $193,980 for the Wind – Diesel configuration. Further analysis solidifies the PV-Diesel HES as the best choice with a payback period of 1.3 years, 75 tons of CO_2 emission reduction, and annual savings of $30,583 compared to the initial diesel energy system.

Keywords

Hybrid energy systems; Sustainable energy transition; HOMER; RETScreen; Payback period; Emission

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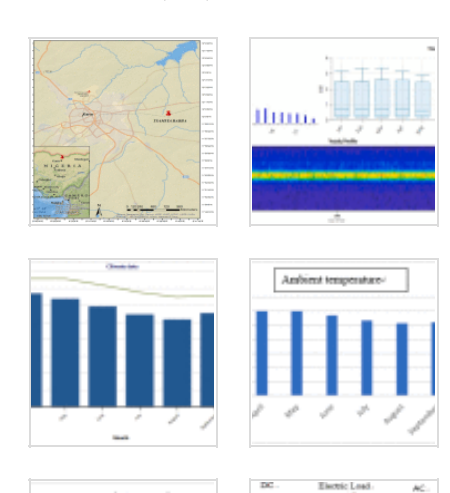

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1. Introduction

The increasing pace of urbanization in countries worldwide is partly due to reliable and sustainable energy supply. Energy is a prerequisite to improving other basic amenities such as healthcare and water supply, which are essential to the daily functioning of human lives [1]. Inadequate energy supply reduces the chance of socio-economic development and results in lower living standards [2]. Many developing countries face this particular problem, especially in Africa, and this is a major cause of the poor socio-economic development on the continent, particularly in the health sector. A prime example of this is Kano State, the second-most populous state in Nigeria. In 2020, during the height of the Covid-19 pandemic, Kano experienced a strong wave of infections that spread from the urban areas to the rural communities [3]. In response, the state government ordered the construction of temporary wards around local hospitals to improve their operational capacity [3,4]. However, no provisions were made for corresponding increases in power supply to these new sections. More often than not, the lone generator used could not handle the increased load demand. Consequently, a vast majority of the new units could not be supplied with electricity, which hampered their intended purpose of catering to those infected with the COVID-19 virus. This trend was consistently observed in Nigeria and many other African countries as they struggled with the subsequent waves of the pandemic.

This deplorable situation resulted in several problems such as substandard sterilization of medical equipment, electricity power failure during surgery, inadequate electricity to support sophisticated medical equipment, etc. [5]. By improving the power supply, the quality of healthcare provided by these clinics can be improved, and this is possible by tapping into the renewable energy potential in these locations. One proven way of achieving this is using hybrid energy systems (HESs). Single-source standalone systems have inherent shortcomings, including high initial cost and intermittent energy production [6]. In contrast, hybrid combinations of renewable energy with conventional fossil fuel sources such as a diesel generator (DG) and a backup storage system have proven less costly and more reliable for electrifying rural communities [7]. Each energy source can be harnessed to its full potential and serves as a backup for the entire system. The performance of these HESs is highly essential to ascertain the specific technical and economic benefits over the single-source standalone system, especially in providing energy to rural health centres.

The National Renewable Energy Laboratory developed HOMER software in the United States in 1993. HOMER can design, evaluate and optimize technical and financial options for renewable energy installations, which may be off-grid, on-grid, or standalone power systems [8]. Halabi et al. utilized HOMER to model two decentralized power stations, each containing a different hybrid PV/Diesel/Battery configuration in Sabah, Malaysia. The authors investigated the impact of different PV penetration levels and the generation of harmful emissions on the system performance. Their study concluded that the performance of standalone non-renewable systems could be greatly improved with renewable energy resources [9]. In this context, Olatomiwa et al. worked on a hybrid energy system for different rural locations in Nigeria; Jos, Sokoto, Iseyin, Maiduguri, Enugu, and Port-Harcourt, analyzing the optimal techno-economic parameters for the provision of electricity to rural health clinics using HOMER software. The result showed that the proposed system was suitable for rural locations near most of these cities [10].

In modelling an energy system, it is necessary to evaluate the environmental impact of such a system. Chowdhury et al. presented an environmentally sustainable hybrid energy system for the health sector in Bangladesh due to the emergence of COVID-19. The author designed and analyzed different configurations of wind/diesel/PV/battery HESs based on emission and cost of electricity using HOMER software. The result showed that the optimized system configuration generated 27% less CO2 than a standalone diesel system [11]. Furthermore, Nag et al. carried out a feasibility and techno-economic analysis of a HES in electrifying a rural location in India. The author designed a cost-effective solar-wind-bioenergy hybrid system to meet the load demand [12]. In addition, a summary of several studies on hybrid renewable energy systems modelling using energy software is presented in Table 1.

Table 1. Summary of different configurations of hybrid renewable energy.

Configuration	System Type	Evaluation criteria	Methodology	Locations
PV/grid [13]	Grid	Tilt angle	HOMER	India
PV/wind/diesel/battery [10]	Off-grid	RF	HOMER	Nigeria
PV/wind/diesel/biomass/battery [14]	Off-grid	NPC, LCOE	HOMER	Bangladesh
Hydro/solar/battery [15]	Off-grid	NPC, GHG	HOMER	Rwanda
PV/grid [16]	Grid	Tilt-angle, LCOE	HOMER	India
PV/grid [17]	Grid	GHG, RF	HOMER	Egypt
P V/diesel/battery [9]	Off-grid	NPC, LCOE	HOMER	Malaysia
P V/wind/diesel/battery [18]	Off-grid	NPC, GHG, LCOE, RF	HOMER	Malaysia
P V/wind/diesel/battery [19]	Off-grid	COE	HOMER	India
PV/grid [20]	Grid	NPC, FS, PR	RETScreen	UAE
PV/biomass [21]	Grid	PR, GHG	RETScreen	Bangladesh
PV [22]	Grid	GHG, NPV	RETScreen	Nigeria
PV [23]	Grid	GHG, NPV, PR	RETScreen	Pakistan
PV [24]	Grid	AS, EP, GHG	RETScreen	Ghana
Solar PV [25]	Grid	GHG, Simple payback, PR	RETScreen	Libya

*Payback period (PR), Renewable fraction (RF), Annual savings (AS), Fuel saving (FS), Equity payback (EP).

Table 1 shows more HESs with different configurations and energy sources, including solar PV, wind, biomass, grid-energy, diesel, and battery. These systems were modeled with either HOMER or RETScreen to conduct techno-economic and environmental analyses using several metrics. Although comprehensive research has been done on utilizing either of the two software used in this project for sizing and optimizing hybrid energy systems, the literature review indicates that each software is being employed solely as both perform similar functions. However, HOMER is more proficient in the technical aspect of the design, while RETScreen is adept in the financial aspect. This project seeks to combine the strengths of the two software packages to produce more robust results, and this trend of combining multiple software in modelling energy systems has seen traction in recent years [26].

Furthermore, many of the abovementioned projects have focused son rural residential communities. Scant research has been conducted on modelling HESs for the healthcare sector, particularly in Africa. Some include Oladigbolu et al. [1], Ani et al. [27], and Olatomiwa et al. [28], who all used HOMER to optimize and design of HESs for healthcare centres in Africa. In this context, we propose a design for an optimal HES with HOMER and RETScreen expert, primarily focusing on rural healthcare centres. The principal aim of this study is to model and design an affordable HES that combines alternative sources of energy with HOMER. The HES's financial sensitivity and risk analysis are performed with the RETScreen Expert. The target location has abundant solar and wind energy potential due to its geographical positioning, and variations of the HES containing one or more of these sources in combination with the DG are investigated.

2. Methodology

2.1. Site description and current energy scenario

The target healthcare facility for this project is Tsamiya Babba rural community healthcare centre in Dambatta local government area (LGA), situated in Kano State, the northern part of Nigeria. The community healthcare is located at 12.4382° N latitude and 8.6161° E longitude, as shown in Fig. 1. Like many rural parts of northern Nigeria, this location is an impoverished region with a vast majority of the population living well below the poverty line. Consequently, the hospital is government-funded and is of modest size. It serves as a general-purpose hospital focusing on children and pregnant women, the most vulnerable in such conditions. The hospital's sections are as follows: reception, emergency ward, consulting room, male and female wards, delivery room, maternity ward, dispensary, records room, and storage room. The electrical appliances used by the hospital are lighting, fans, and refrigerators, as shown in Table 2. Due to its remoteness, the community is not connected to the electricity grid. Thus, the power supply for the facility is a single diesel generator. Furthermore, because of the prevailing economic conditions, this generator is only switched on when needed and primarily to charge appliances or during emergencies as there is limited funding available for the hospital.

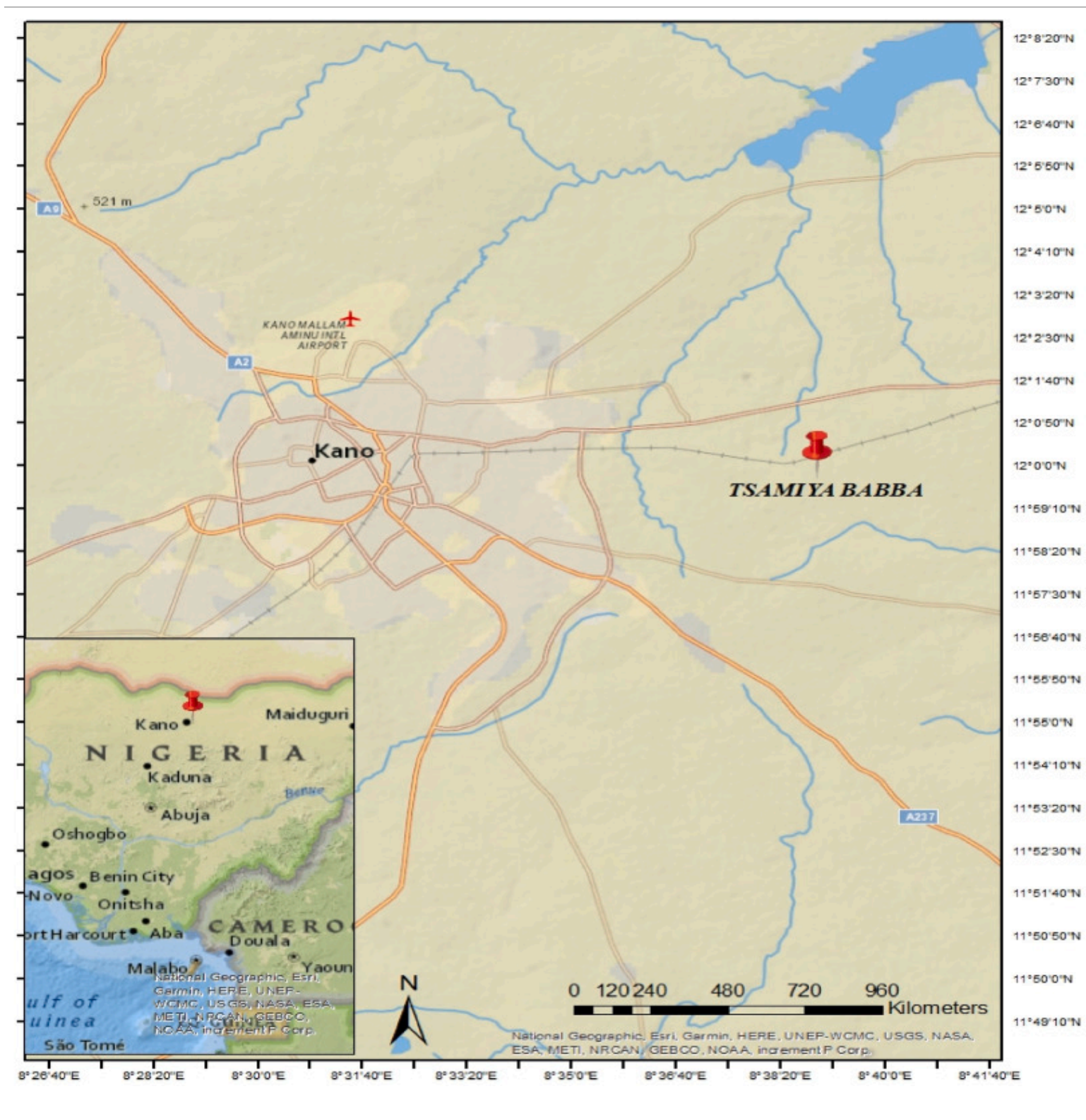
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Fig. 1. The location of Tsamiya Babba village.

Table 2. Electrical load description of the health care centre.

Appliance type	No of Appliance	Rating (W)	Running time/day	Total kWh/day
refrigerator non-medical	1	150	7	1.05
refrigerator vaccine	1	60	12	0.72
desktop computer	1	150	5	0.75
CFL outdoor	6	40	8	1.92
LED lamp	8	18	8	1.152
centrifuge	1	242	3	0.726
Microscope	2	20	6	0.24
blood chemical analyzer	1	45	5	0.225
Hematology analyzer	1	200	4	0.8
CD4 Machine	1	200	4	0.8
Radio	2	15	10	0.3
TV set	1	120	9	1.08
ceiling fan	4	65	7	1.82
Mobile charger	5	18	6	0.54
Ultrasound Machine	1	800	2	1.6
Oxygen Concentrator	1	220	2	0.44
blood bank refrigerator	1	70	18	1.26

Typically, the load in the hospital is maintained at a minimum during the night using rechargeable LED lamps from 12 a.m. to 5 a.m. Then it experiences peak hours from 10 a.m. to 1 p.m. as the staff resume work and the community's inhabitants arrive for consultations. The generator is sometimes turned on to serve the increased demand. Subsequently, the load demand gradually decreases throughout the rest of the day. The daily load profile is shown in Fig. 2. The daily load demand and peak power are 15.8 kWh/day and 2.45 kW, respectively, with a load factor of 0.2. To obtain a better maximum load and due to the particular nature of the facility's load demand, a time-step of 10% and day-to-day random variability of 5% were used for the simulation in HOMER.

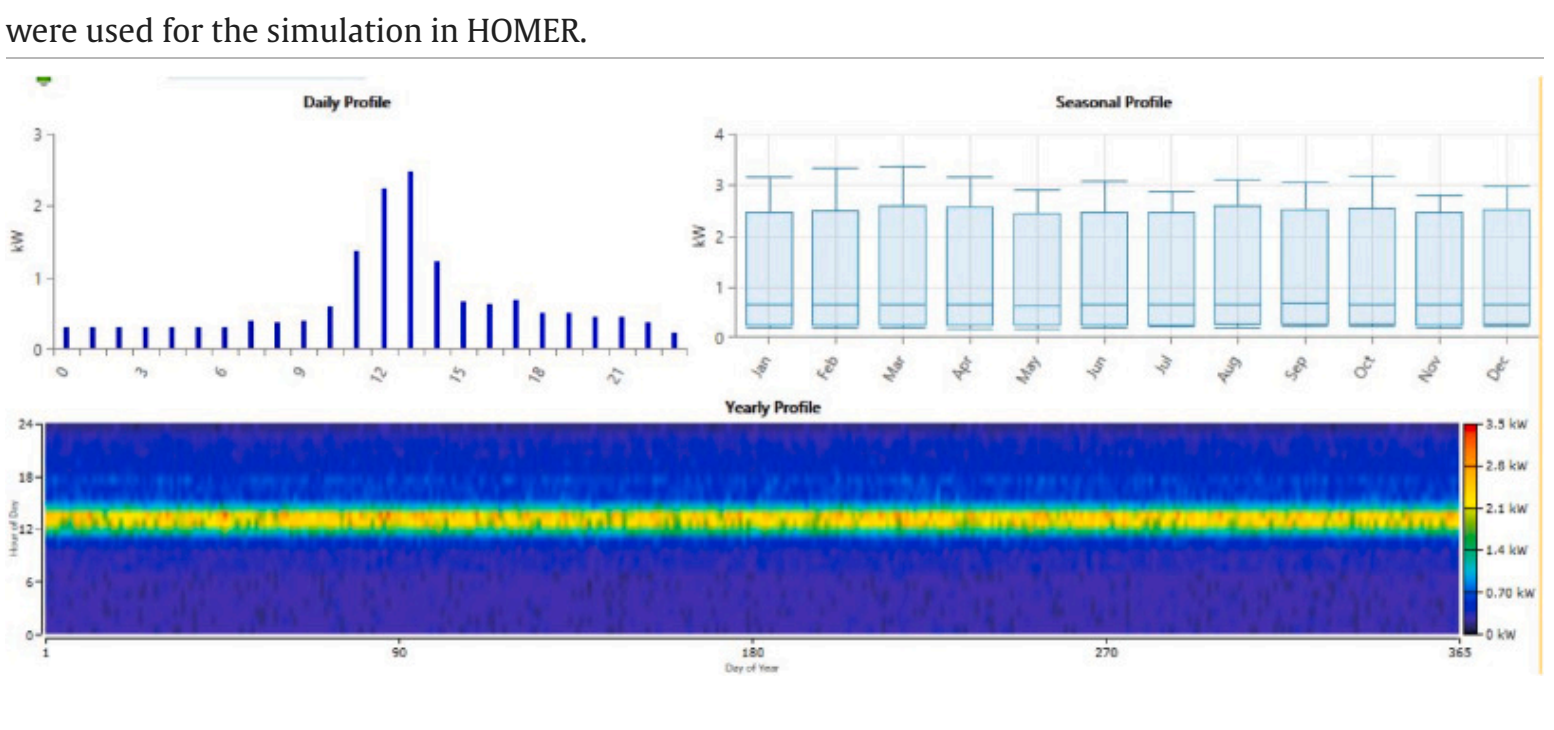

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Fig. 2. The daily load profile of the rural healthcare centre.

2.2. Weather data location

The climatic data utilized for this project was obtained using RETScreen software. RETScreen possesses an inbuilt NASA meteorological database, which can be accessed by specifying the longitude and latitude of the desired location. The parameters influencing energy production from renewable energy sources available at the target hospital were obtained with this dataset. The two most impactful variables for solar energy are the mean solar irradiation per month and the mean ambient temperature per month. It can be observed that the mean solar irradiation is 5.86 kWh/m^2/day, as shown in Fig. 3. The temperature varies from 30 °C to 20.3 °C, with an average temperature of 25.7 °C, as shown in Fig. 4. Regarding wind energy, the critical factor is the average wind speed of 3.4 m/s at the target location, as shown in Fig. 5. When this data was inputted into HOMER for optimization, it generated artificial hourly time-series data that mimic the characteristics of actual wind speed data.

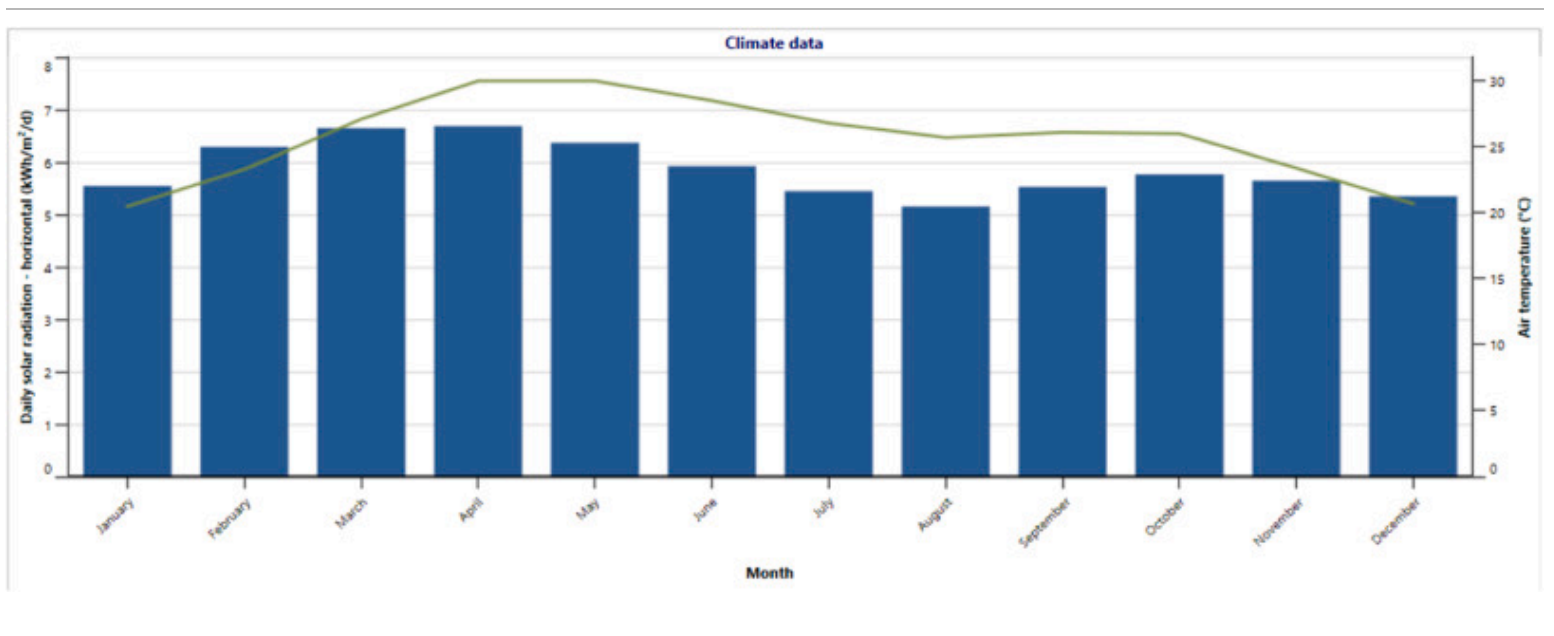

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Fig. 3. The monthly mean solar irradiation of Tsamiya Babba rural healthcare centre.

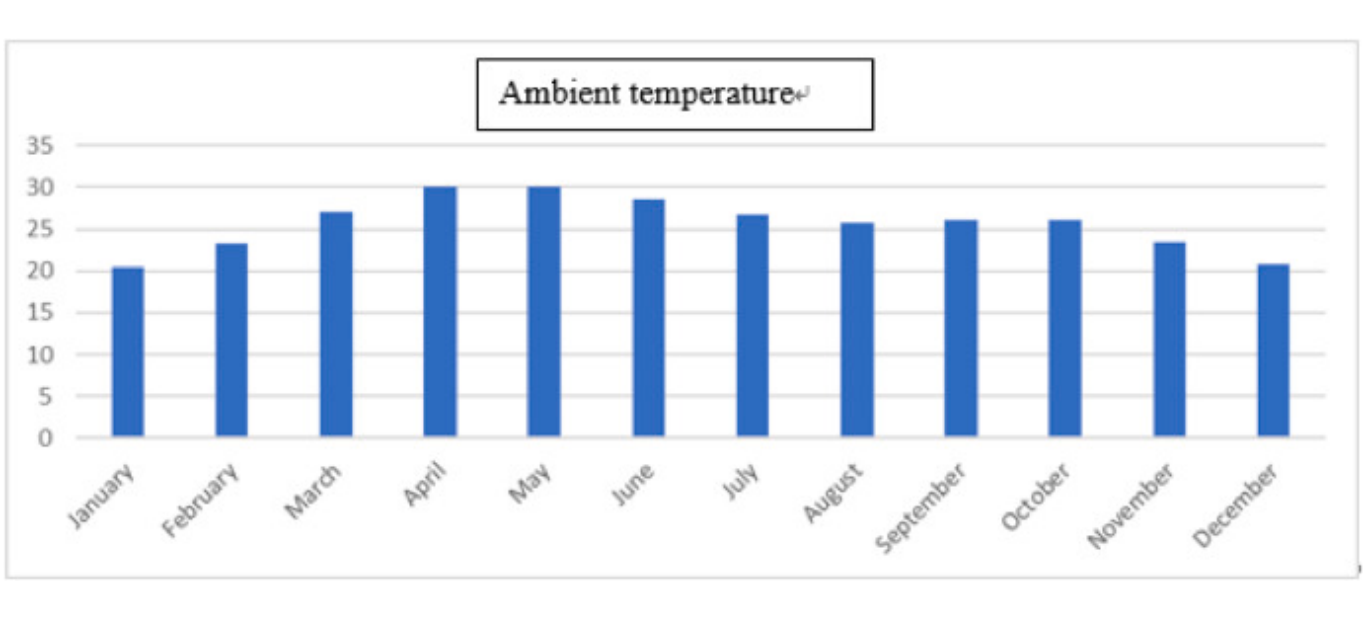

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Fig. 4. The monthly mean ambient temperature of Tsamiya Babba rural healthcare centre.

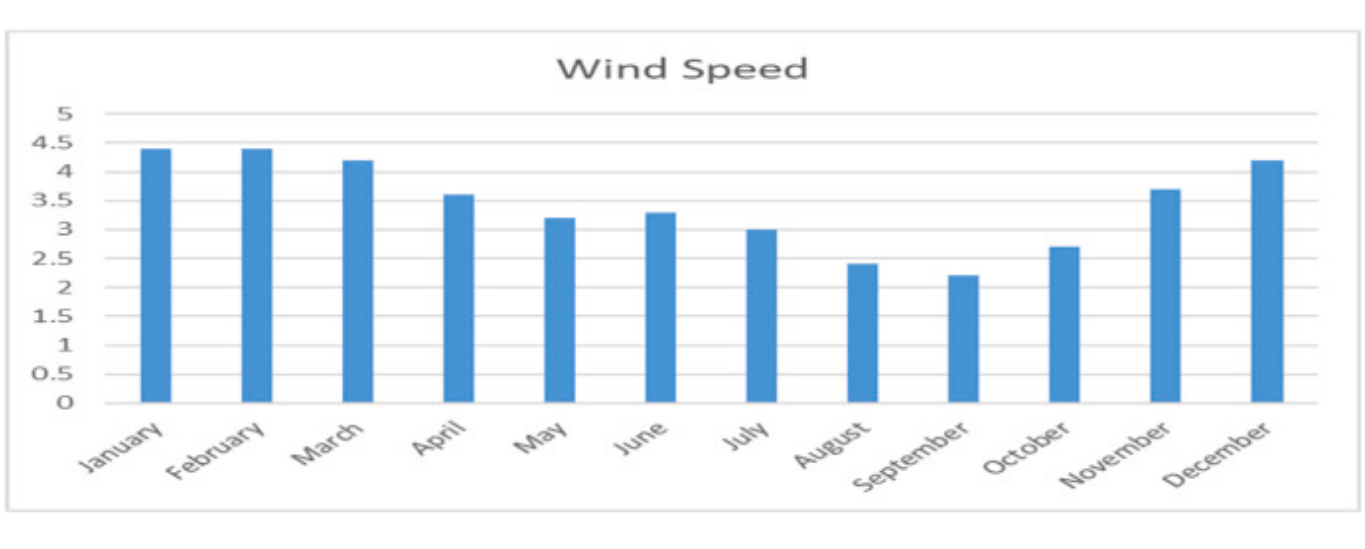

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Fig. 5. The monthly average wind speed of Tsamiya Babba rural healthcare centre.

2.3. Modelling HES components

The proposed HES is comprised of a diesel generator, solar PV array, wind turbine, converter, and batteries. Each component is modeled with several parameters within HOMER to produce accurate results.

2.4. Diesel generator

Diesel generators (DG) provide a continuous source of electricity in a HES, given the availability of fuel. They are typically used as backup when other intermittent sources operate below optimal level or as a complementary energy source in peak consumption hours. DGs are characterized by their minimum loading ratio, usually about 25–30% [29]. This is the lowest portion of the rated power that the DG can operate at to prevent low speed and poor fuel combustion. The cost variables for DGs are heavily influenced by their fuel intake (in L/h), which is given by equation (1) [1,29]:

$$C_{DG} = c_f \cdot T_{DG} + m \cdot P_{DG} \tag{1}$$

where c_f is the fuel intercept coefficient in L/kWh, T_{DG} is the generator's capacity in kW, m is the fuel slope in L/kWh, and P_{DG} denotes the diesel generator's output.

2.5. Solar PV array

Photovoltaic modules produce DC power by converting solar irradiation into electrical energy. A collection of such modules is referred to as an array. The power produced by a PV array is heavily dependent on the amount of solar irradiation present at the installation site and its ambient temperature. The PV module selected for this study is the mono-Si – CSUN200-72 M module manufactured by China Sunergy. The nominal operating cell temperature (NOCT) and the temperature coefficient of the panel are specified as 45 °C and 0.4% °C, respectively, as shown in Table 3. The maximum power produced by a single panel is given by Refs. [5,22]:

$$P_{MP} = \frac{G_i}{1000} \times P_{MP,STC} \times [(T - 25)\,\delta_{MP} + 1] \tag{2}$$

$$T = T_{AMB} + (NOCT - 20) \times \frac{G_i}{800} \tag{3}$$

where G_i is the incident solar irradiation at the site in W/m^2, $P_{MP,\,STC}$ is the maximum power produced by the module at STC conditions in W, δ_{MP} is the temperature coefficient, T is the module's actual temperature, and T_{AMB} is the ambient temperature in °C which are given in the equation above.

Table 3. China Sunergy solar module parameters.

Property	Values
Maximum Power (P_{max})	200 W
positive Power Tolerance	0–3%
Short Circuit Current (Isc)	5.72 A
Open Circuit Voltage (V_{OC})	45.3 V
Maximum Power Voltage (Vmpp)	37.6 V
Maximum Power Current (Impp)	5.32 A
Efficiency	15.67%
Nominal operating temperature	45 °C
Temperature coefficient	0.4% °C

2.6. Wind turbine

Wind turbines produce energy by converting the kinetic energy of moving wind into mechanical energy with their blades (rotors) and then to AC electric power via a generator. The productivity of a wind turbine is highly influenced by the available wind speed and the rotor swept area of the turbine. For this study, a generic wind turbine was employed on HOMER, while a Bergey Windpower (BERGEY BWC EXCEL – 18.3 m) with a hub height of 18.3 m and a rating of 30 kW was chosen for RETScreen. The mechanical and electrical power of a wind turbine is given by the following equations (4), (5) [1,30].

$$P_M = \frac{1}{2} \times \rho \times A \times V^3 \tag{4}$$

$$P_E = \frac{1}{2} \times \rho \times A \times V^3 \times C_P \times 10^{-3} \tag{5}$$

where ρ is the air density (1.225 kg/m^3), A is the rotor swept area in m^2, V is the wind speed in m/s, and C_P is the power coefficient of the wind turbine. Table 4 gives the property of the wind turbine employed.

Table 4. Wind turbine parameters.

Property	Values
Manufacturer	Bergey Windpower
Model	BERGEY BWC EXCEL-18.3 m
Rated capacity	10 kW
Hub height	18.3 m
Rotor diameter	7 m
Swept area	38.48 m

2.7. Converter

In a hybrid system with different energy sources producing AC and DC power, a bi-directional converter is used to switch from one form to the other. When transforming from AC to DC, the converter operates as a rectifier, and from DC to AC, it functions as an inverter. Converters typically have a high level of efficiency (around 95–99%) for both inverting and rectifying modes, low maintenance, and a relatively long lifetime. In this study, a 23 kW converter was utilized. The power capacity of the converter is given using equation (6):

$$C = 3 \times (L_I + L_R) \tag{6}$$

where L_I is the total inductive load and L_R denotes the total resistive loads.

2.8. Battery bank

A battery bank is mainly used to complement other energy sources in the HES, especially the non-dispatchable sources such as wind and solar. The batteries store excess energy when production is more than consumption and discharge it when the opposite is true. The two major battery types commercially available are lead-acid batteries and lithium-ion batteries. The operating lifetime of the batteries depends on the depth of discharge (DOD), which is the maximum portion of power that can be drawn from the battery in one cycle without adversely affecting it. The battery chosen for this study is a 12 V 1230Ah lithium-ion battery with a depth of discharge of 0.6 because it is readily available in the Nigerian market, and its efficiency is 0.85% [22]. The nominal capacity of the battery bank needed is given by Ref. [22]:

$$E_U = E_{max} - E_{min} \tag{7}$$

$$E_N = \frac{E_U \times A}{DOD_{max} \times \eta_i \times \eta_b} \tag{8}$$

where E_U is the useful daily energy, E_{max} is the maximum daily energy stored in the battery bank, E_{min} is the minimum daily energy discharged, A is the days of autonomy, DOD_{max} is the depth of discharge, η_i is the converter's inverting efficiency, and η_b is the battery bank's efficiency. The cost associated with the various system components is presented in Table 5.

Table 5. Cost associated with different components.

Component	Initial cost	Replacement cost	Maintenance cost	Reference
PV system	$1800/kW	$1800/kW	$18/year	[22]
Wind turbine	$4000/kW	$3200/kW	$200/year	[1]
Diesel generator	$200/kW	$200/kW	$0.05/kW/h	[1]
Converter	$200/kW	$200/kW	–	[31]
Battery	$174/unit	$174/unit	$5/unit	[10]

2.9. System constraints and dispatch strategy

Besides the modelling parameters, several constraints also need to be imposed when using HOMER to reflect practical realities [10]. Constraint parameters such as operating reserve, capacity shortage, and percentage of renewable output were set to 10%, 0%, 80%, and 20% (solar and wind), respectively. Furthermore, a hybrid renewable energy system that comprises diesel generators and batteries requires a dispatch strategy [1]. The dispatch strategy employed for this study is the cycle charging dispatch strategy, in which the diesel generator charges the storage system and meets its current load demand. This strategy is employed to extend the battery's lifespan and reduce the cost of operation of the diesel engine because the diesel generator is only used when the renewable energy source and the battery system cannot meet the load demand.

3. Results and discussion

To obtain the optimal configuration, HOMER was used to optimize different HES configurations. Then, RETScreen was used to conduct a techno-economic analysis on them. Considering the available resources at the target location, two different configurations were modeled to supply power to the centre. These include:

1. Solar PV-Diesel system.
2. Wind-Diesel system.

3.1. Simulation of solar PV-diesel system with HOMER

The peak load of the healthcare centre is 21.5 kW, and the parameters used for modelling the system in HOMER are 5% maximum annual capacity shortage, 10% operating reserve of the load, 10% annual interest rate, and project life of 25 years. The configuration of the system design is shown in Fig. 6. In this system, 22 kW of PV generates 85.5% of the total supply for a total value of 37,573 kWh/year at a capacity factor of 20%. The generator produces 6373 kWh/year, which is equivalent to 14.5% of the total supply. The system has a fuel consumption of 1910 L/year and a run time of 4451 h/year. The system produced excess electricity of 3,320 kWh/year. The renewable fraction, the percentage of energy generated from renewable sources, is 83.1%. The monthly electricity production of the design is shown in Fig. 7. The system design and specific costs associated with each component, such as total net present cost, capital, operation and maintenance, and fuel cost, are given in Table 6, Table 7.

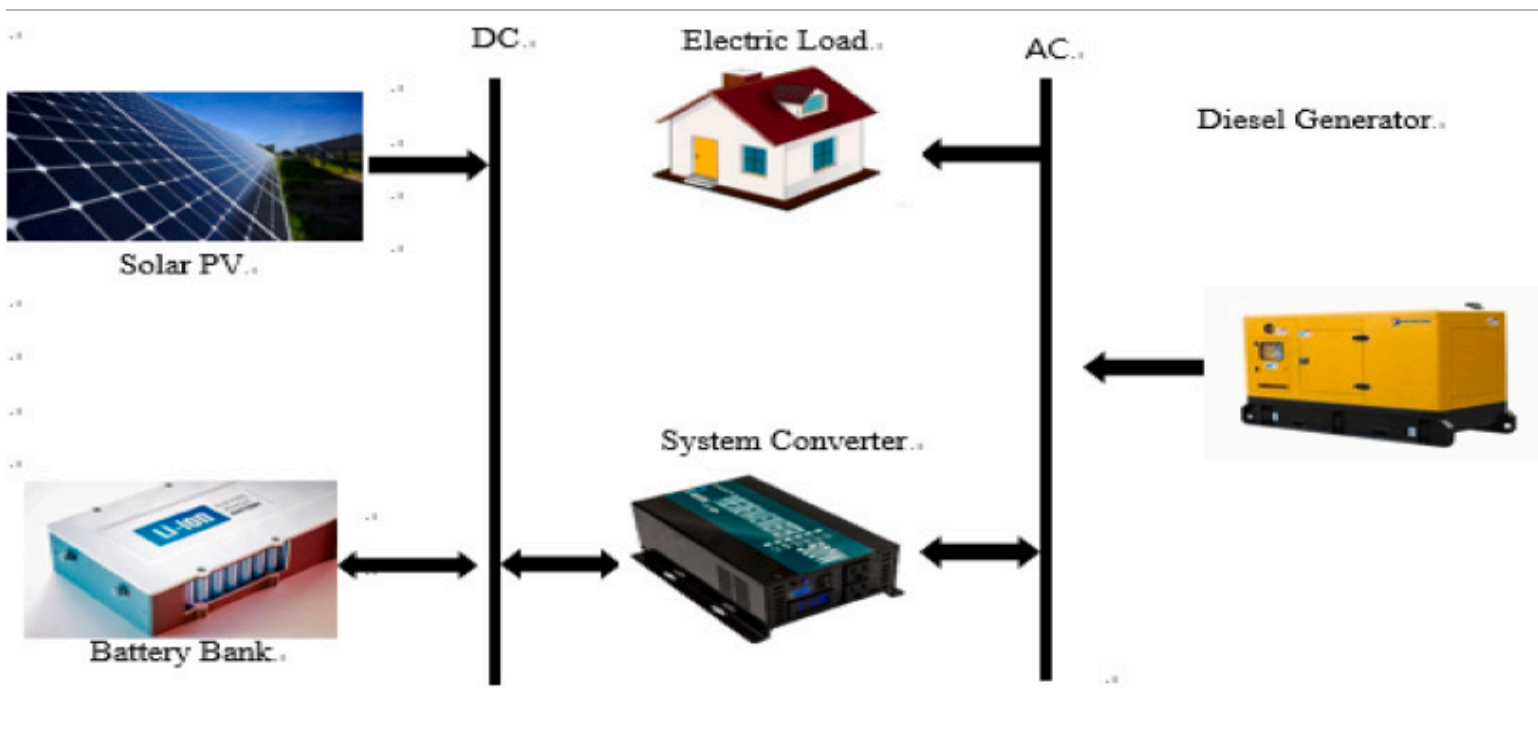

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Fig. 6. The system configuration solar PV-Diesel design.

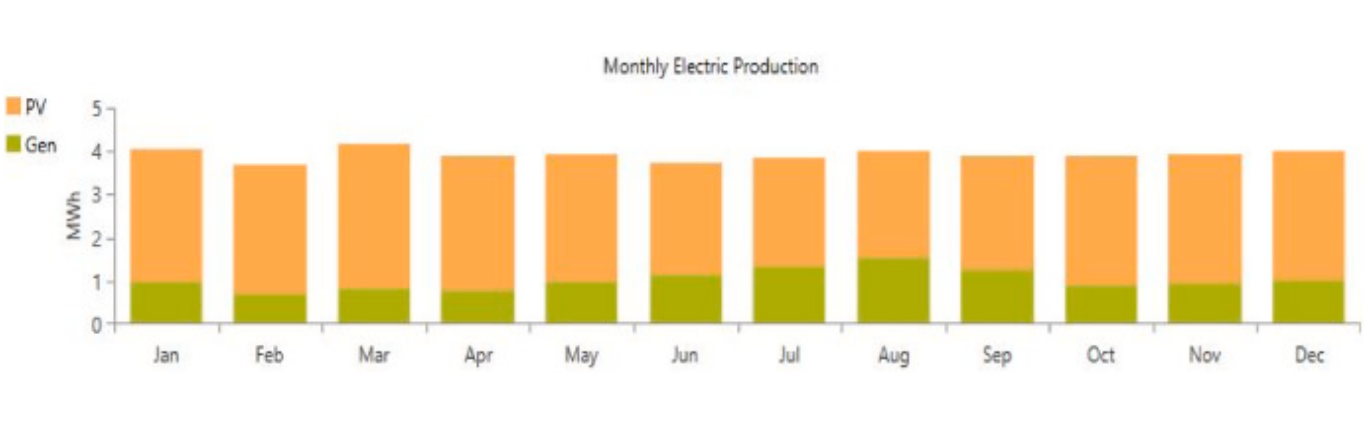

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Fig. 7. Monthly electricity production of the PV-Diesel design.

Table 6. The system parameters for the PV-Diesel system.

PV	Diesel gen.	Battery	Converter	COE	Total net present cost	Operating cost
22 kW	27 kW	198	23.9 kW	$0.28	$134,770	$2318.56

Table 7. The specific cost associated with each component of PV-Diesel.

Component	Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV	39,600	39,600	396	0	0	79,596
Diesel Gen.	5400	5400	318	15,088.78	772	26,978.78
Battery	34,452	34,452	990	0	401.36	70,295.36
Converter	4780	4780	0	0	471.86	10,031.86
System	84,232	84,232	1704	15,088.78	1645.22	186,902

A single battery has a nominal capacity of 125 kWh.

3.2. Simulation of solar Wind-Diesel system with HOMER

In this system, day-to-day and time step random variability were set to 5% and 10%, respectively, to match the peak load of 21.5 kW, and the configuration of the system is shown in Fig. 8. A 36 kW of wind system generates 69.7% of the total supply for 34,053 kWh/year at a capacity factor of 10.8%. Meanwhile, the Diesel generator produces 30.3%, which equates to 14,806 kWh/year at a capacity factor of 6.3%. The system has a fuel consumption of 6391 L/year and a run time of 6338 h/year. The system produced excess electricity of 8935 kWh/year with a renewable fraction of 60.7%. The monthly electricity production of this system is shown in Fig. 9. The system design and specific costs associated with each component, such as total net present cost, capital, operation and maintenance, and fuel cost, are given in Table 8, Table 9.

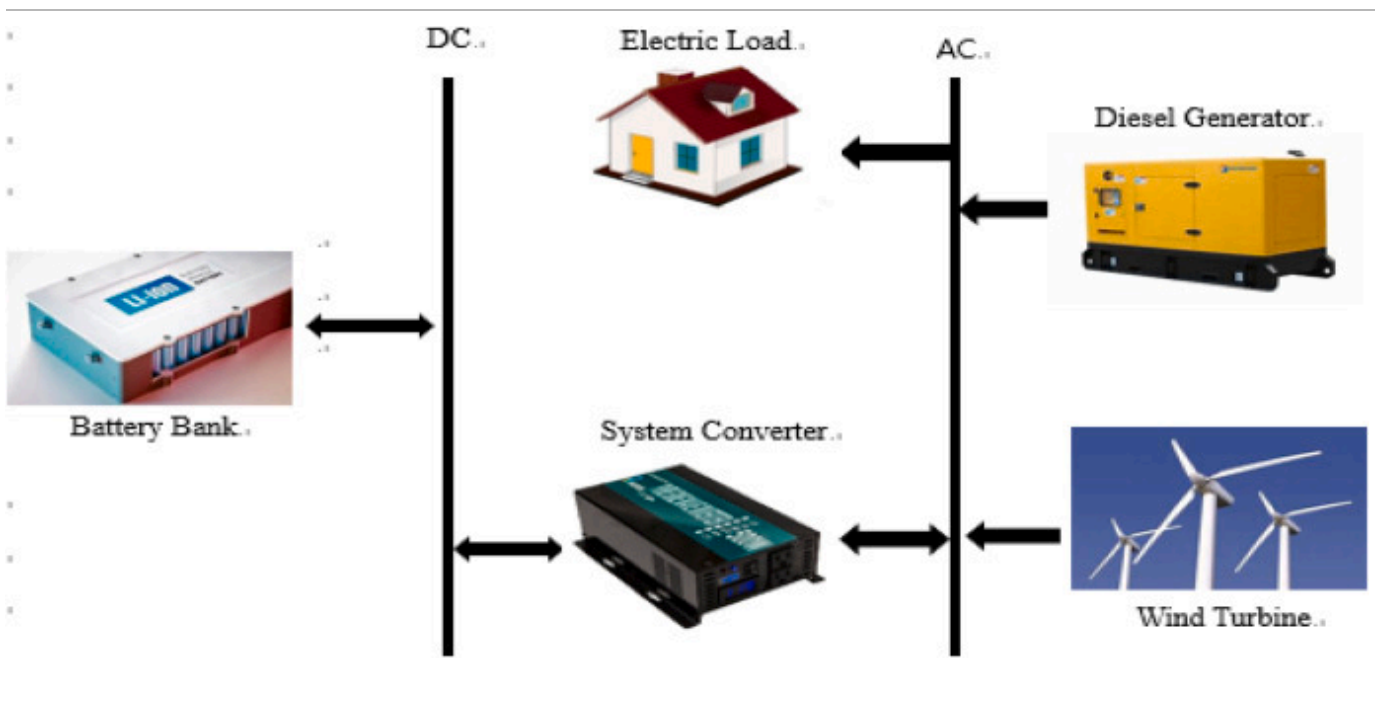

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Fig. 8. The system configuration solar PV-Diesel design.

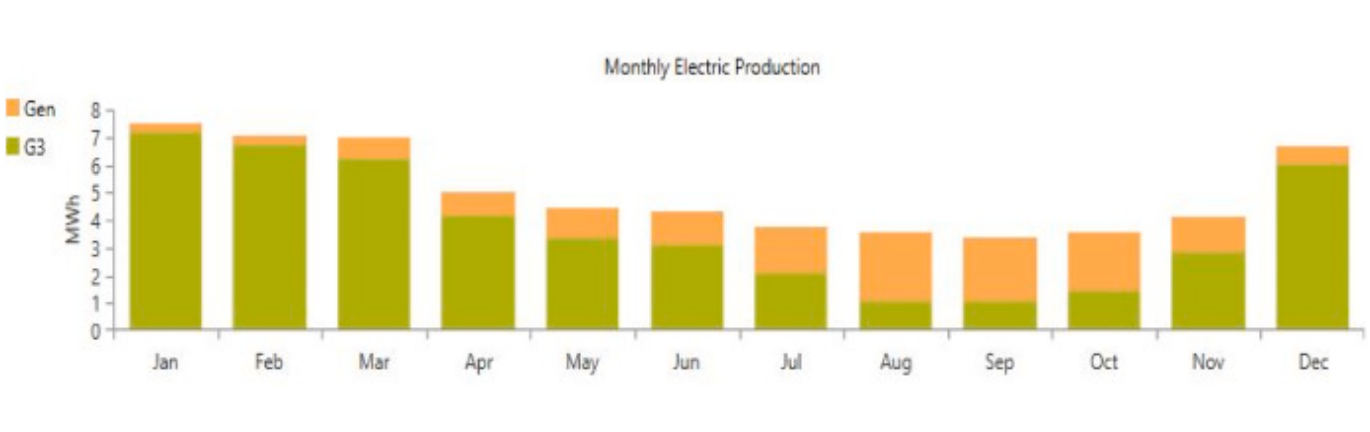

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Fig. 9. Monthly electricity production of the Wind-Diesel design.

Table 8. The system design for the Wind-Diesel system.

Wind turbine	Diesel gen.	Battery	Converter	COE	Total net present cost	Operating cost
36 kW	27 kW	396 units	20.6 kW	$0.28	$268,745.70	$12,424.44

Table 9. The specific cost associated with each component of case 2.

Component	Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
Wind turbine	60,000	0	40,906.74	0	0	104,906.74
Diesel Gen.	5400	7046.10	20,139.054	50,877.31	954.13	82,508.81
Battery	30,821.47	29,980.11	12,655.52	0	0	73,074
Converter	6067.72	2531.85	0	0	471.26	8128.31
System	109,919.72	39,558.06	73,829.63	50,877.31	5439.02	268,745.70

A single battery has a nominal capacity of 125 kWh

3.3. Simulation with RETScreen

RETScreen is a state-of-the-art software developed by the Government of Canada to empower low-carbon and renewable energy decisions. It provides functionalities for identifying, modelling, and optimizing renewable energy, energy efficiency, and co-generation projects [22]. The viability and feasibility of this work were examined with this software. After obtaining the optimized result from HOMER, the results were inputted into RETScreen to conduct the feasibility analysis, as shown in Fig. 10. Additional costs such as engineering, feasibility, and development costs were included and were given 18.2%, 1.8%, and 5% of the total initial cost, respectively [26]. Since both designs included a sizable portion of renewable energy, RETScreen also estimated the greenhouse gases (GHG) reduced by the system. Hybrid energy systems with renewable energy sources are inherently intermittent. Factors such as temperature, wind speed, and hours of sunshine must be considered in designing such a system. Furthermore, other factors such as local infrastructure, financial viability, reliability, and environmental impact were considered to determine the optimum renewable fraction.

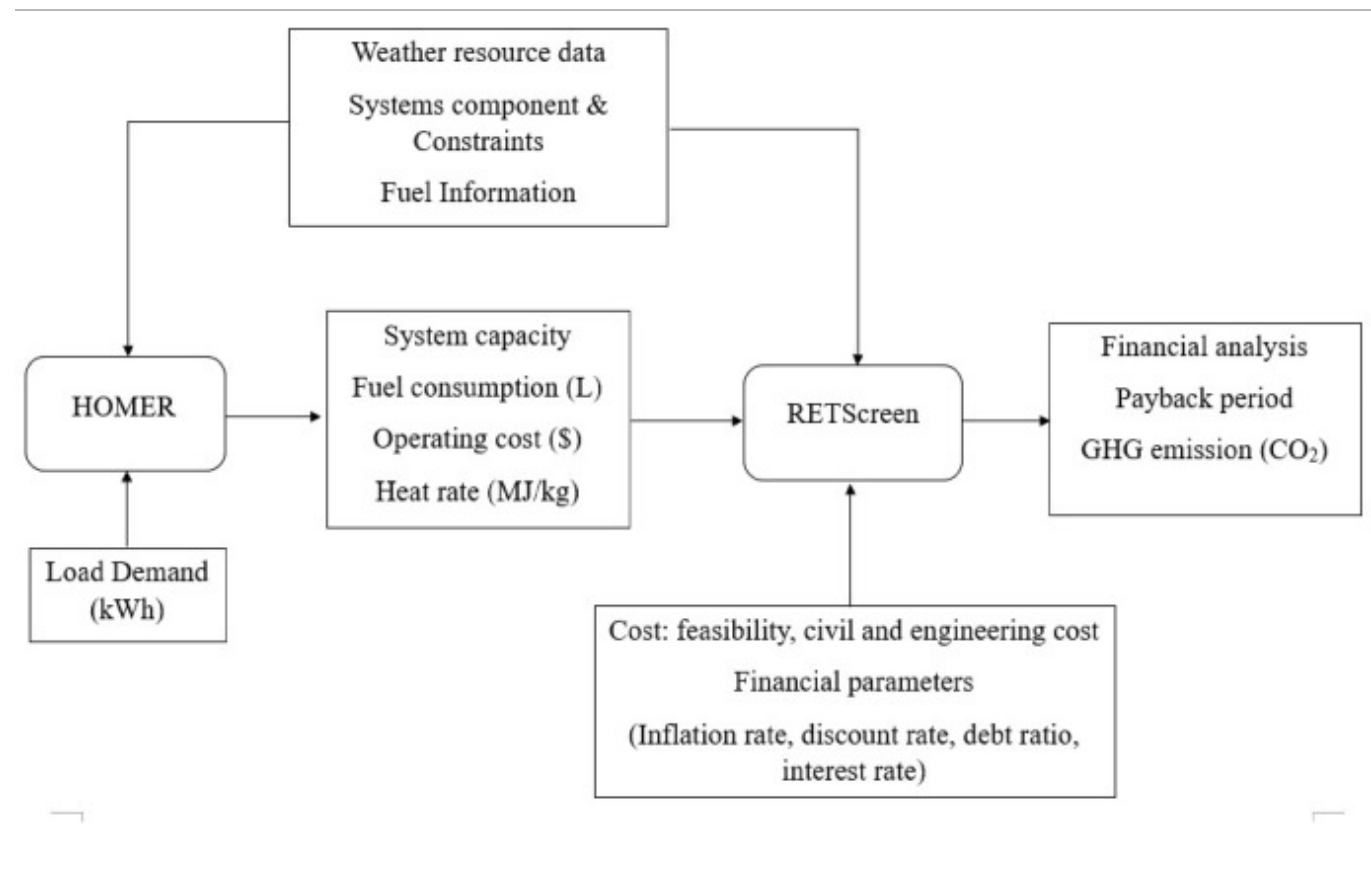

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Fig. 10. Overall system methodology process.

Feasibility studies must be carried out before commencing any project to have an overview of the sustainability and viability of the project. Using RETScreen software, the return on investment, payback period, risk analysis, net present value, equity payback, benefit-cost ratio, total annual cost, and GHG emissions were obtained. Equity payback is the number of years it takes to recover the total invested capital. Therefore, a viable project should have a short equity payback period. Since the initial source of electricity for the healthcare centre is a Diesel generator (base case), this study seeks to compare the proposed designs (Wind-Diesel and PV-Diesel systems) with the base case. The estimated debt rate, inflation rate, discount rate, debt interest rate, and project lifetime were 70%, 2%, 6%, 7%, and 25 years for the proposed design.

3.3.1. Financial analysis of the project with RETScreen

After obtaining the optimized result of both proposed designs from HOMER, the results obtained were used as input variables in RETScreen software to get the financial clarity of the project. The parameters inputted include a peak load of 21.5 kW, a fuel rate of 0.619 $/L, 1910 L/year as the fuel quantity, and a load demand of 103 kWh. One hundred fifteen units of China Sunergy mono-Si – CSUN200-72 M with a capacity factor of 20% equivalent were used to generate 30.4 MWh of solar PV. The inputs for case 2 are a peak load of 21.5 kW, fuel rate of 0.619 $/L, 6391 L/year as the quantity of fuel consumed, and the load demand of 103 kWh. Four units of Bergey Windpower turbines (BERGEY BWC EXCEL – 18.3 m) with a capacity factor of 7.2% generated 18.8 MWh of electricity. For the financial viability of these systems, input variables such as escalation rate fuel, inflation rate, discount rate, reinvestment rate, debt ratio, debt interest rate, debt term were pre-determined [22], excluding the total annual cost, initial cost, operation and maintenance (O&M) cost as presented in Table 10. The simulation result from RETScreen showed the annual life savings, the internal rate of return (IRR), the net present value (NPV), benefit-cost (B–C) ratio, and debt service charge, as presented in Table 11. The financial viability is determined by its payback period, IRR, and net present value [22]. The Solar PV – Diesel system has annual savings of $30,583 with an NPV of $390,949 compared to $15,174 and $193,980 for the Wind–Diesel system. In addition, the simple payback and equity payback periods are 3.2 and 1.3 years for solar and 10.7 and 10.6 years for wind, as shown in Fig. 11. All these indicate that both cases are economically viable and make financial sense irrespective of the choice. Still, it can be observed that solar has a higher net present cost and the shortest payback period compared to wind. This implies that the photovoltaic system will save the healthcare centre the cost of running only a Diesel generator for 27.3 years. Therefore, it is best to implement this configuration for the healthcare centre financially.

Table 10. Financial variables from RETScreen software.

Financial inputs	PV-Diesel	Wind-Diesel
Escalation rate fuel	2%	2%
Inflation rate	2%	2%
Discount rate	6%	6%
Reinvestment rate	9%	9%
Project life	25yr.	25yr.
Debt ratio	70%	70%
Debt interest rate	2%	2%
Debt term	15yr.	15yr.
Debt payment	$7916	$34,789
O &M cost	$3425	$5527
Total Initial cost	$150,100	$452,648

Table 11. Financial output parameter of the project.

Financial viability	PV-Diesel	Wind-Diesel
Internal rate of return	82.1%	13.1%
Net present value ($)	$390,949	$193,980
Annual life cycle savings ($/year.)	$30,583	$15,174
Benefit-cost (B–C) ratio	9.7%	2.4%
Debt service coverage	4.2%	1.2%

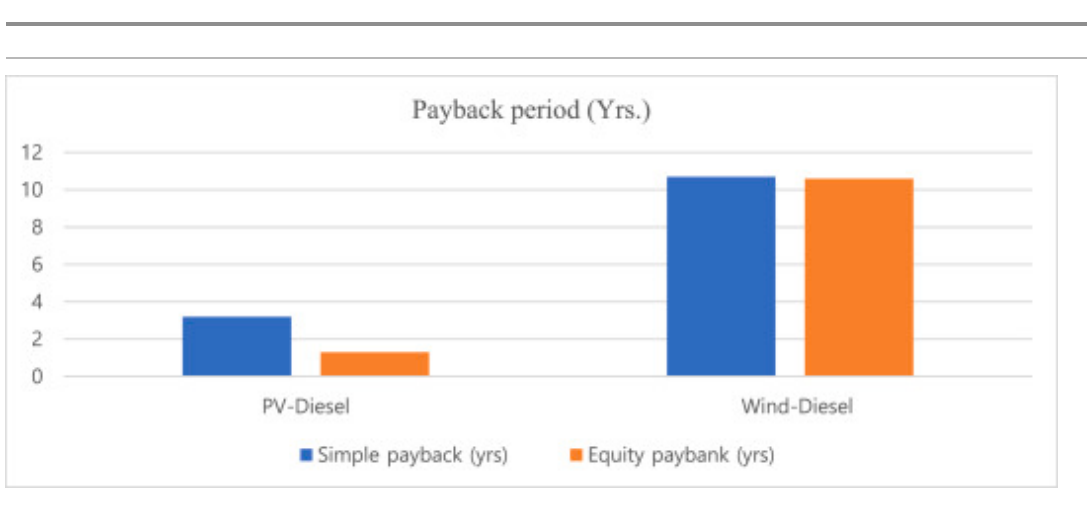

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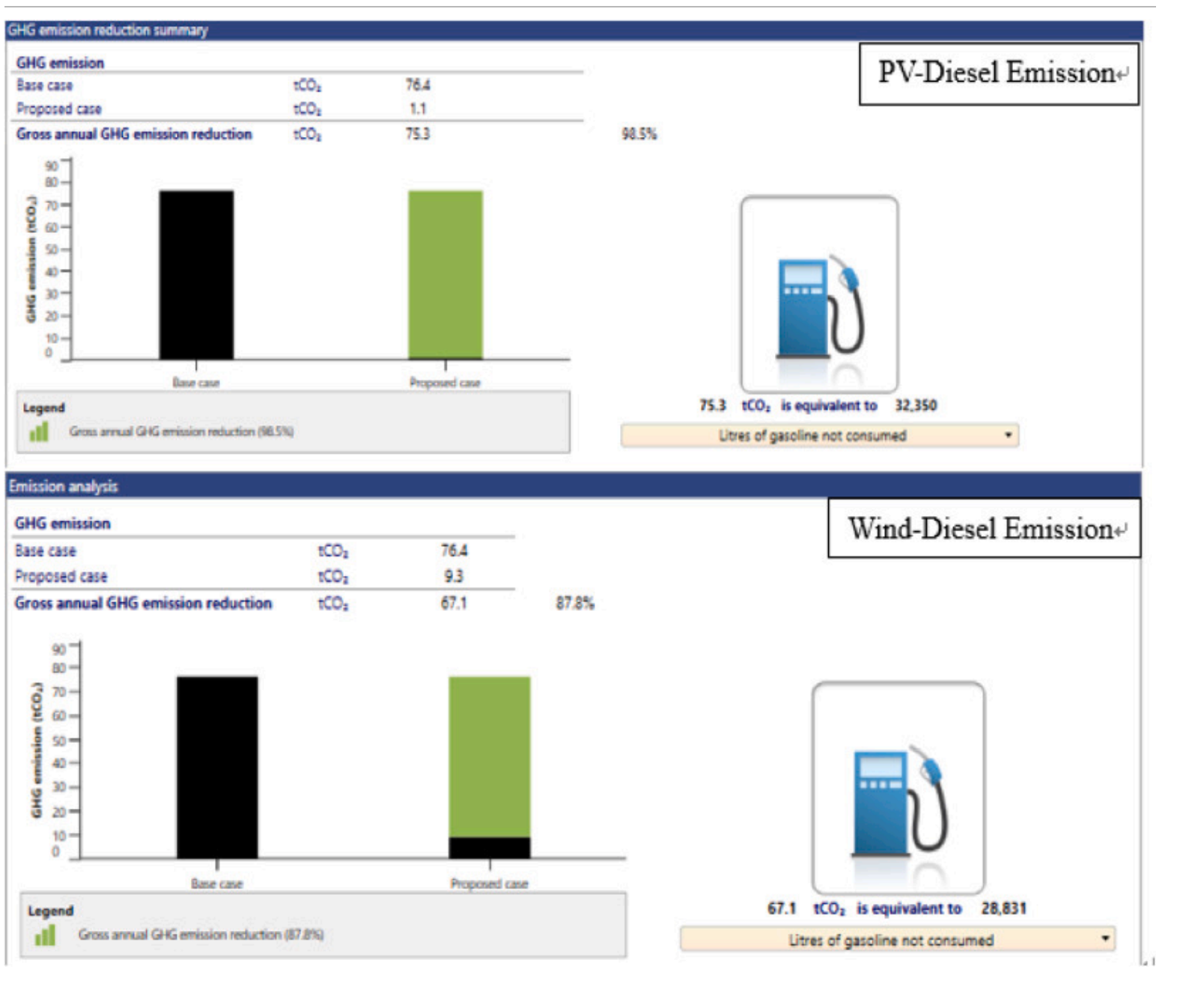
Fig. 11. The payback period of the project.

3.3.2. Emission reduction assessment

It is crucial to consider the economic viability of renewable or hybrid renewable energy systems and their environmental influence. Implementing a system that lowers greenhouse gas (GHG) emissions will positively reduce environmental pollution and global warming. In this study, RETScreen has calculated the gross annual GHG emission reduction of the proposed cases compared to the base case of diesel standalone, as shown in Fig. 12. The difference between the emission of the base case and proposed cases gives the gross annual GHG emission. The PV-Diesel system has the maximum gross annual GHG emission reduction of 75.3 tons of CO_2, equivalent to 32,350 L of gasoline not consumed yearly. In addition, the Wind-Diesel system has a GHG emission reduction of 67.1 tons of CO_2, equal to 28,831 L of gasoline not consumed. This shows that the PV-Diesel system is economically viable and contributes the most to emission reduction. Hence, supporting the United Nations Framework Convention on Climate Change (UNFCCC) on low carbon emission development strategies.


GHG emission reduction summary

PV-Diesel Emission

GHG emission			
Base case	tCO_2	76.4	
Proposed case	tCO_2	1.1	
Gross annual GHG emission reduction	tCO_2	75.3	98.5%

75.3 tCO_2 is equivalent to 32,350 Litres of gasoline not consumed

Emission analysis

Wind-Diesel Emission

GHG emission			
Base case	tCO_2	76.4	
Proposed case	tCO_2	9.3	
Gross annual GHG emission reduction	tCO_2	67.1	87.8%

67.1 tCO_2 is equivalent to 28,831 Litres of gasoline not consumed

GHG emission (tCO₂)
90
80
70
60
50
40
30
20
10
0
Base case
Proposed case
Legend
Gross annual GHG emission reduction (98.5%)
Gross annual GHG emission reduction (87.8%)

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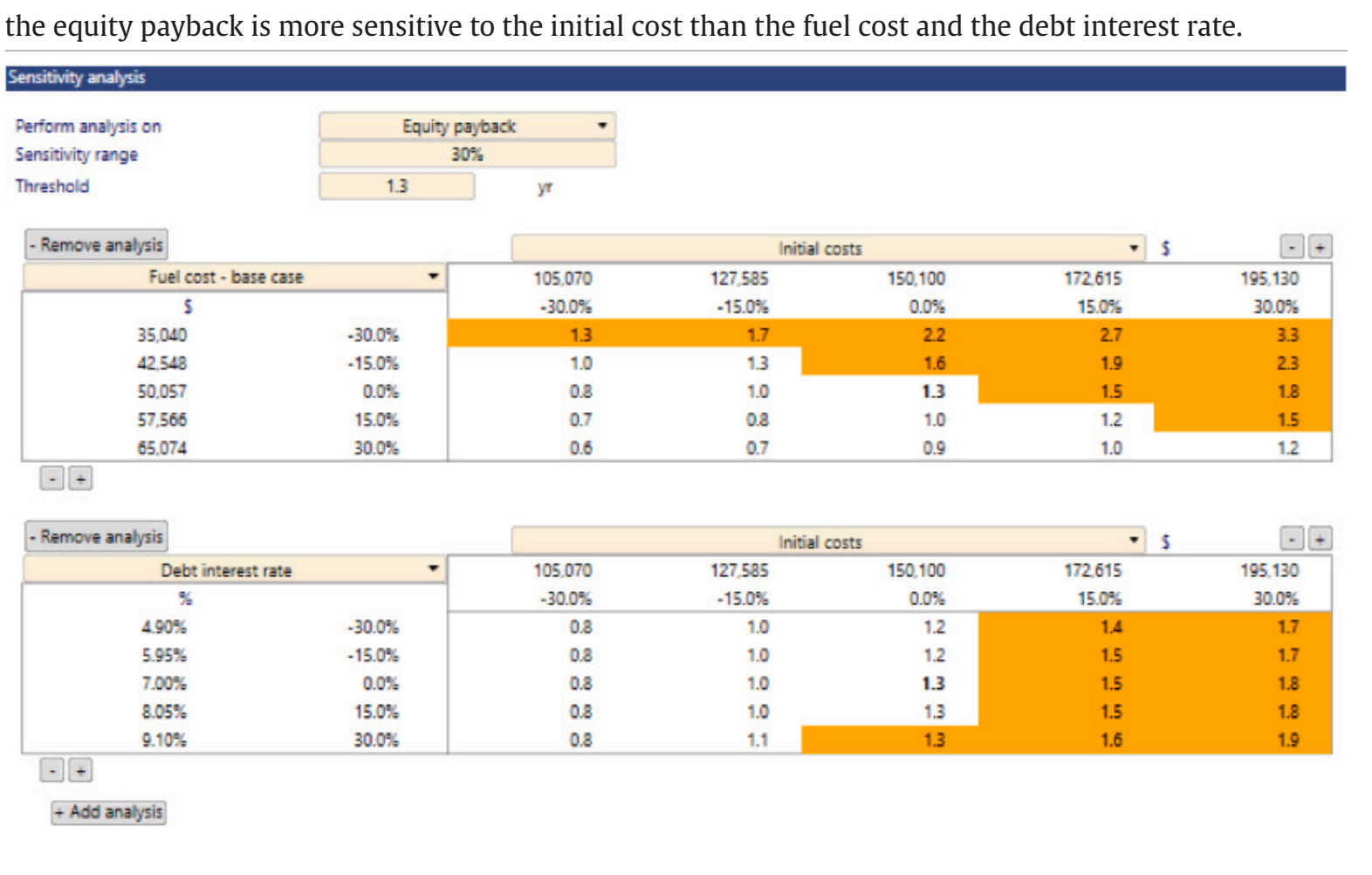
Fig. 12. The greenhouse gas emission of the systems.

3.3.3. Sensitivity analysis

Sensitivity analysis is the variation of different parameters such as fuel cost, debt term, payback period, and debt interest rate associated with the system to evaluate their effects on the calculated financial variables. This analysis can reduce the level of financial uncertainty based on inputted parameters. This analysis was only carried out for the PV-Diesel system because it has the best financial and environmental viability. The sensitivity analysis was evaluated for the equity payback period of the project by iterating the fuel cost of the base case against the Initial cost and the debt interest rate against the initial cost by a sensitivity range of ±30% and a threshold of 1.3 years as shown in Fig. 13. The fuel cost was varied at ± 30% against the initial cost of $150,100, resulting in $105,070 and $195,130. In addition, the initial interest rate is 7%, and with a ±30% variation gives 9.10% and 4.9%, respectively. RETScreen specifies the value of the equity payback lower than the stated bound-limit of 1.3 years. If the project's initial cost increases by 30% and the fuel cost decreases by the same amount, the project will not be financially profitable because the equity payback period will be 3.3 years. If the fuel cost increases and the initial cost is reduced, the project would be economically viable because the equity payback will be 0.6 years. This shows that the equity payback is more sensitive to the initial cost than the fuel cost of the project. Furthermore, when the initial cost varies against the debt interest rate, the result shows that the project will only be financially viable if the initial cost decreases and the debt interest rate increases. In both sensitivity cases, it is observed that the equity payback is more sensitive to the initial cost than the fuel cost and the debt interest rate.


Sensitivity analysis

Perform analysis on: Equity payback
Sensitivity range: 30%
Threshold: 1.3 yr

Fuel cost - base case		Initial costs $				
		105,070	127,585	150,100	172,615	195,130
$		-30.0%	-15.0%	0.0%	15.0%	30.0%
35,040	-30.0%	1.3	1.7	2.2	2.7	3.3
42,548	-15.0%	1.0	1.3	1.6	1.9	2.3
50,057	0.0%	0.8	1.0	1.3	1.5	1.8
57,566	15.0%	0.7	0.8	1.0	1.2	1.5
65,074	30.0%	0.6	0.7	0.9	1.0	1.2

Debt interest rate		Initial costs $				
		105,070	127,585	150,100	172,615	195,130
%		-30.0%	-15.0%	0.0%	15.0%	30.0%
4.90%	-30.0%	0.8	1.0	1.2	1.4	1.7
5.95%	-15.0%	0.8	1.0	1.2	1.5	1.7
7.00%	0.0%	0.8	1.0	1.3	1.5	1.8
8.05%	15.0%	0.8	1.0	1.3	1.5	1.8
9.10%	30.0%	0.8	1.1	1.3	1.6	1.9

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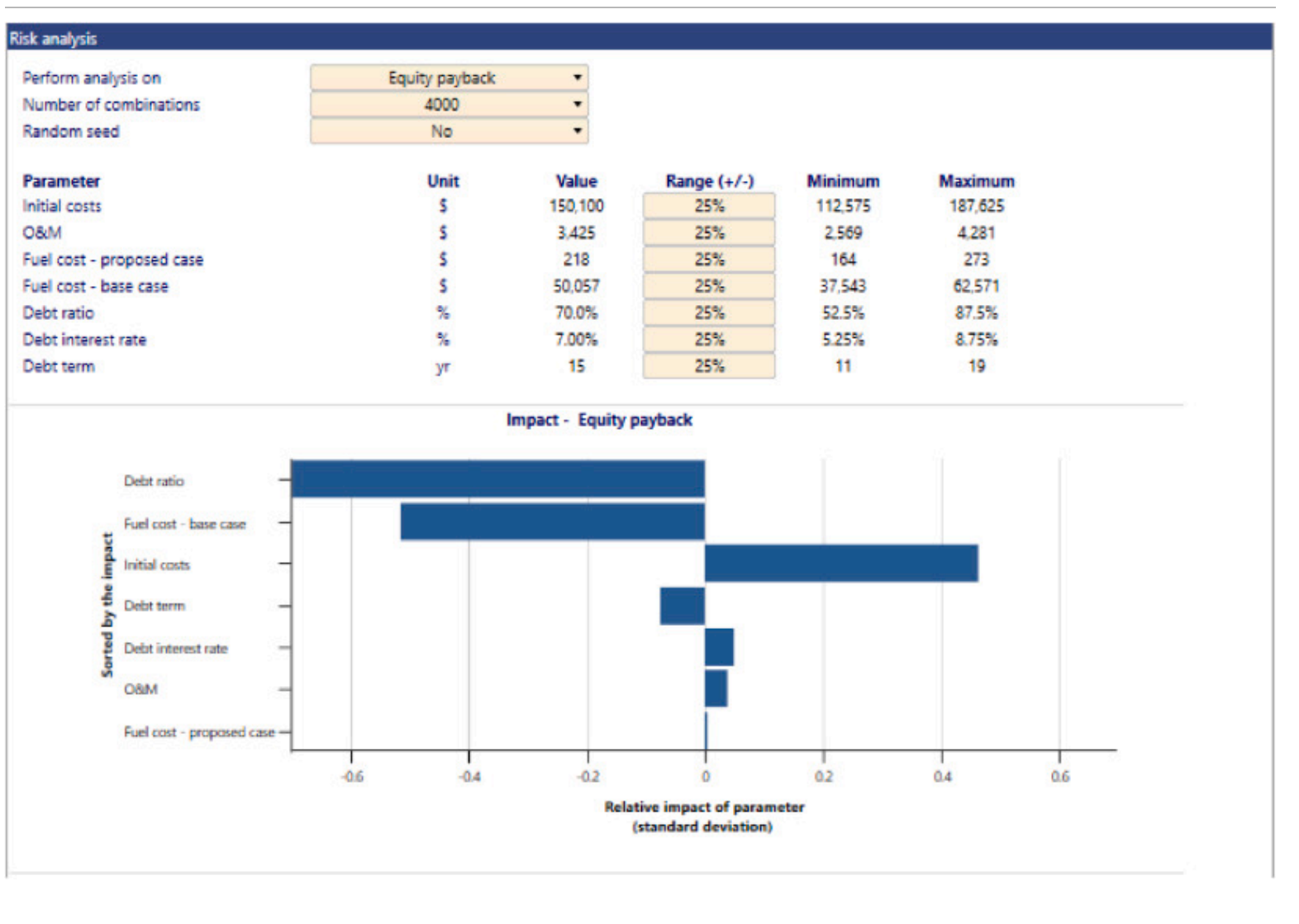
Fig. 13. Sensitivity analysis result for the PV-Diesel system.

3.3.4. Risk analysis

Unlike sensitivity analysis, the risk analysis allows varying more than two parameters associated with the project within a specific range. This can help the project advisor reach a financially grounded conclusion. The equity payback was chosen as the financial indicator, and a range of ±25% variation factor was selected for all parameters. RETScreen software was used to carry out risk analysis for the PV-Diesel project using the Monte Carlo simulation techniques for 4000 iterations [22] for re-estimating the equity payback period. Fig. 14 shows that the equity payback was obtained by varying several parameters, as shown by the impact graph. The debt ratio and the initial cost have a huge impact on the project but the opposite. An increase in the initial cost increases the equity payback period, while an increase in the debt ratio decreases equity payback. Similarly, an increase in fuel cost (base case) decreases the equity payback, while an increase in debt interest rate increases the equity payback.


Risk analysis

Perform analysis on: Equity payback
Number of combinations: 4000
Random seed: No

Parameter	Unit	Value	Range (+/-)	Minimum	Maximum
Initial costs	$	150,100	25%	112,575	187,625
O&M	$	3,425	25%	2,569	4,281
Fuel cost - proposed case	$	218	25%	164	273
Fuel cost - base case	$	50,057	25%	37,543	62,571
Debt ratio	%	70.0%	25%	52.5%	87.5%
Debt interest rate	%	7.00%	25%	5.25%	8.75%
Debt term	yr	15	25%	11	19

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Fig. 14. Risk analysis Impact graph of the PV-Diesel system.

4. Conclusion

The Covid-19 pandemic has seen a lot of collective effort applied toward healthcare and vaccination. However, these medical facilities also need adequate power to function, and this is sorely lacking on the African continent, especially in rural areas. Therefore, this project has conducted a comprehensive study to model, simulate and analyze different configurations of a hybrid energy system (HES) for a rural healthcare centre in northern Nigeria. Two configurations of such a HES, PV – Diesel and Wind – Diesel, were modeled and simulated using HOMER to obtain each design's performance and capital costs. Then RETScreen Expert was used to execute financial, emission, risk, and sensitivity analysis on the chosen system. For a PV-Diesel HES system, the initial cost of this system amounted to $150,100, and its payback period is found to be 1.3 years, with an annual gross GHG emission reduction of 75.3 tCO_2.

Furthermore, the system has an annual net savings of $30,583 compared to the base case. Whereas for the Wind-Diesel system, the equity payback is 10.6 years with an initial cost of $452,683 and an annual gross GHG emission reduction of 67.1 tons of CO_2. It also has an annual net savings of $15,174 compared with the base case. Considering the renewable resource potential at the target facility location, the PV-Diesel HES system is the most suitable for the healthcare Centre, as evidenced by the calculated financial metrics. Further analysis also showed that the PV-Diesel system is the most suitable option from the emission, operation, and maintenance perspective.

5. Future recommendations

This paper investigates different configurations of a HES to determine the optimum choice for a rural healthcare centre. However, future study needs to consider the viability of grid extension to manage the excess energy. In addition, computational techniques such as genetic algorithm (GA) and grouped method of data handling (GMDH) can also be applied in optimizing and analyzing potential HES designs.

Credit author statement

Abdulfatai Olatunji Yakub: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. Noel Ngando Same: Investigation, Data curation. Abdulhameed Babatunde Owolabi: Resources, Writing - review & editing. Benyoh Emmanuel Kigha Nsafon: Revision. Dongjun Suh: Software, Validation. Jeung-Soo Huh: Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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> …Results show that solar-diesel HRES can save about $30,583 per year, with a net present value of $390,949, a 1.3-year payback period, a 75-ton CO2 emission reduction, and $30,583 in annual savings over the initial diesel system. An inherent drawback of the study conducted by [48] is the consideration of a single energy storage technology. The inclusion of battery and hydrogen storage technologies could significantly enhance the reliability and dispatchability of the HES, particularly during periods of low renewable energy availability.…

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